BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: *[Smart Door Lock System using Arduino and ESP32-CAM]*

by

[Group_No: 2]
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Group Members:

Name	ID
Pranto Roy	22301261
Fariha Tanha	22101296
Nishat Tasnim	22301643
Shamaila Sadat Niha	22301728

Department of Computer Science and Engineering
BRAC University

5/5/2026

Abstract

Our project introduces an intelligent door lock with Arduino and ESP32-CAM. The system integrates three forms of access control: password-based access with a keypad, face recognition with ESP32-CAM and remote control with Telegram. To open and close the door, the Arduino uses the keypad, LCD and servo motor to open the door and close it. The ESP32-CAM will recognize faces and signal the unlocking of the door when a known face is sensed. It also enables users to take photos and have control over the lock remotely using Telegram. The system enhances security as it gives several authentication strategies and live monitoring. It can be used in intelligent homes and lockup systems.

Keywords

Arduino, ESP32-CAM, Face Recognition, IoT, Smart Door Lock

1. Introduction

The conventional door locks apply keys or simple passwords that are not very secure. Losing keys, guessing passwords is possible. Our project aims to create a smart door lock system with Arduino and ESP32-CAM which can offer greater security by providing several authentication options. This project is applicable to Computer Interfacing as it is going to connect hardware (keypad, LCD, servo) with software and communication systems (WiFi, Telegram, camera).

2. Related Work/ Inspiration

Most door lock systems utilize just a RFID or a keypad. Others sophisticated systems employ fingerprint sensors. Most systems however, do not integrate several security options to a single system. Our project is unique in that it integrates:

- Password system
- Face recognition
- Remote Telegram control This ensures that the system is more secure and flexible than the traditional systems.

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3. Technical Approach

- **System Architecture:**

  The system consists of two major components,

1. Arduino System:

- Password input keypad.
- LCD for display
- Servo motor to lock/unlock.
- Responds to ESP32.

2. ESP32-CAM System:

- Face recognition
- Telegram communication
- Signals Arduino unlock.

Both systems give messages via digital pins.

- **Components Used:**

Arduino UNO, ESP32-CAM (2 modules), 4x4 Keypad, LCD (I2C), Servo Motor, Transistor, 10k Resistor, Hall sensor, Jumper wires, Breadboard.

- **Cost Breakdown:**

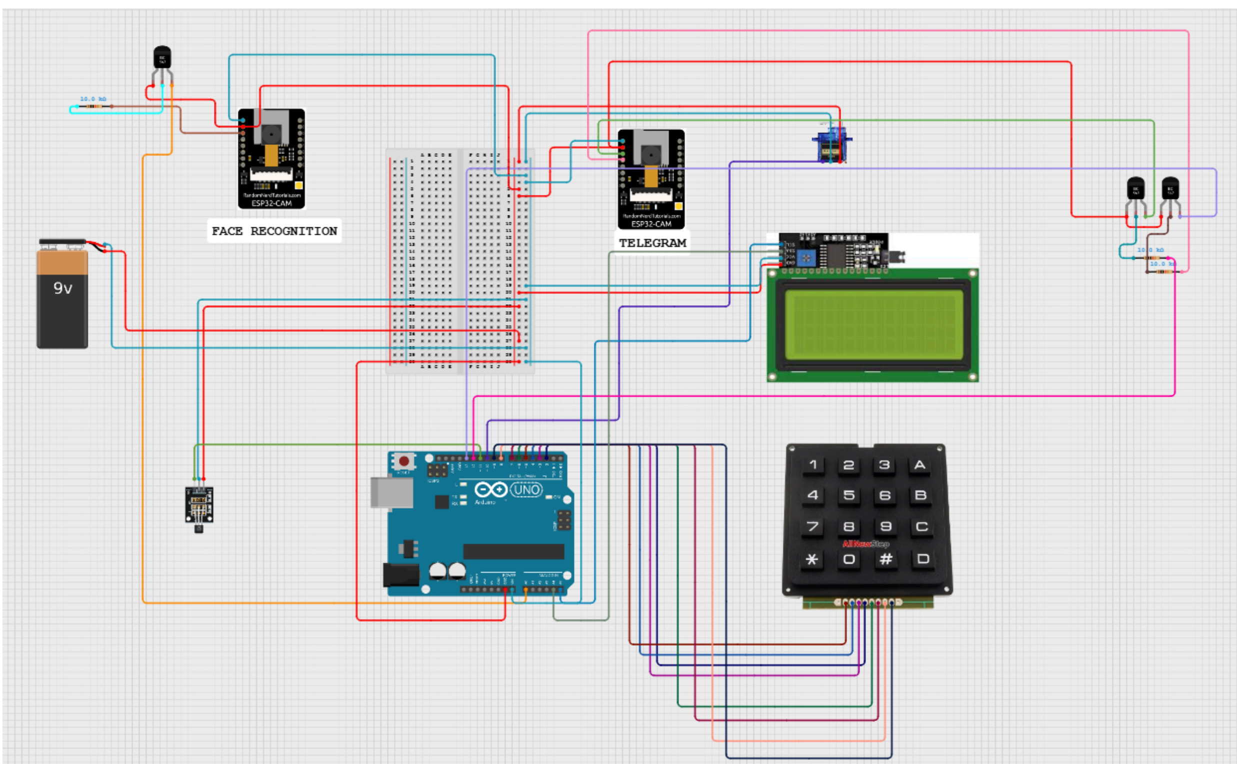
Item	Qty	Unit Price (৳)	Total (৳)
ESP32 CAM	3	900	2,700
Arduino UNO	1	900	900
I2C 20x4 Display	1	600	600
4x4 Keypad	1	350	350
Male Jumper	2	150	300
Servo	1	120	120
Hall Effect Sensor	1	90	90
Breadboard	2	120	240
PCB Sheet	1	350	350
Super Glue	3	20	60
Power Supply 3A	1	550	550
Power Supply 2A	1	450	450
Servo Motor	1	120	120
Power Cable	1	80	80

LED	1	90	90
			7000/=

- **Communication Protocol:**

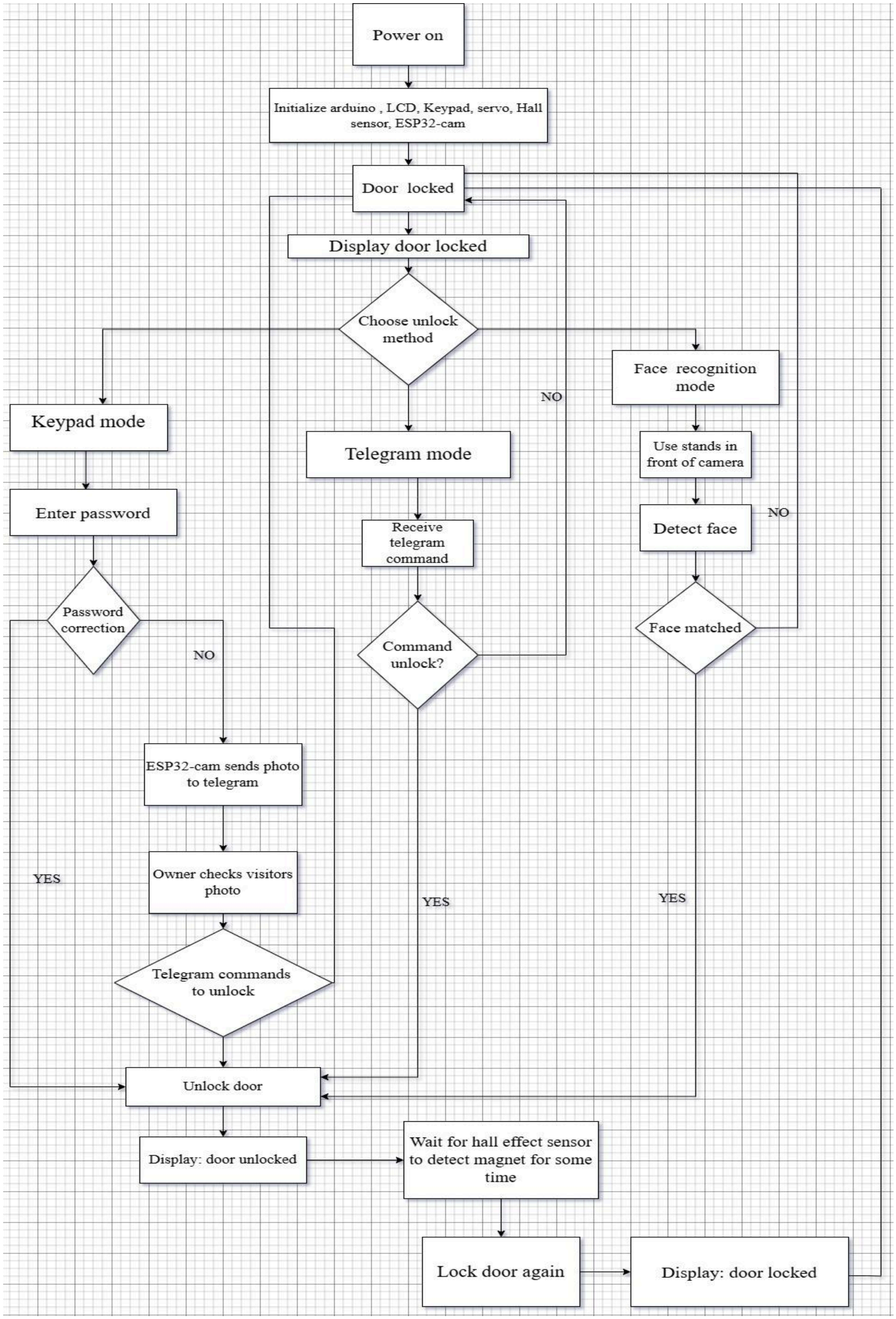
UART: used in debugging and output of serial monitors.
I2C: I2C is used in order to communicate between Arduino and LCD.
WiFi: Used by ESP32-CAM to connect to the internet and hotspot.
HTTP: It is used as a camera web server.
HTTPS: It is employed to provide Telegram communication that is secure.
GPIO: It is used in signaling the transaction between ESP32 and Arduino.
PWM: This is used to control the servo motor.

- **Circuit/Schematic:**



Circuit diagram link

- **Implementation:** (Flow Diagram)


Power on
Initialize arduino , LCD, Keypad, servo, Hall sensor, ESP32-cam
Door locked
Display door locked
Choose unlock method
Keypad mode
Telegram mode
Face recognition mode
Enter password
Password correction
NO
YES
ESP32-cam sends photo to telegram
Owner checks visitors photo
Telegram commands to unlock
Receive telegram command
Command unlock?
NO
YES
Use stands in front of camera
Detect face
NO
Face matched
YES
Unlock door
Display: door unlocked
Wait for hall effect sensor to detect magnet for some time
Lock door again
Display: door locked

- **Working Procedure:**

The smart door lock system is designed with: Arduino UNO, ESP32-CAM module, 4x4 keypad, 20x4 I2C LCD, micro servo motor, hall effect sensor, and transistor switching circuits. The system is also designed to ensure that the door is locked by default and unlocked with a valid access method.

-The Arduino UNO and ESP32-CAM modules automatically begin to run when the system is powered on. Arduino programs the keypad, LCD, servo motor, and hall effect sensor. The ESP32-CAM modules boot up the camera and wireless communication capabilities. Initially, the position of the servo motor is adjusted to the locked position and the LCD displays the message, Door Locked. Once the system has been initialized, it is awaiting an unlock request.

-The system offers three ways (unlocking) which include: keypad password, Telegram command and face recognition. The keypad mode involves entering the password in the 4x4 keypad. The Arduino reads the input of the keypad by reading the row and column pins. Once the password has been typed in, the Arduino will compare it to the stored password in the code. When the password is correct the Arduino will rotate the servo motor to the unlock position and the LCD will display Doors unlocked. In case of the wrong password, the door is locked and LCD displays “Wrong Password.”

-In case of a wrong password, an Arduino sends a trigger signal to the ESP32-CAM module. Once this signal is received, the ESP32-CAM takes a photo of the person in the vicinity of the door and transmits it to the owner via Telegram. The owner is then able to view the photo of the visitor and make a decision as to whether or not he can unlock the door remotely. In case the owner would like to give access, the owner sends the unlock command via Telegram.

-The ESP32-CAM is connected to WiFi and communicates with Telegram via the Telegram Bot API, in Telegram mode. The commands that the owner can send include: /photo, /unlock, /lock and /start. The /photo command allows taking a photograph and transferring it to Telegram. The unlock command sends an unlock signal (via the transistor interface) between the ESP32-CAM and the Arduino. This signal is sent to the Arduino which unlocks the door by using the servo motor. The “/lock” command keeps the door locked or returns the servo motor to the locked position.

-In mode of face recognition, the user is placed before the camera of ESP32-CAM. The ESP32-CAM takes the snapshot and verifies both the detected face with the stored face data. In case the face is a match, the ESP32-CAM uses its GPIO output pin to send a HIGH signal to the Arduino. This signal is read by the Arduino which then opens the door with the help of the servo motor. The LCD, in turn, displays Door Unlocked. A non match between the face and the match does not send any signal to unlock, therefore the door does not unlock.

-The last step in unlocking the door is carried out by the Arduino, whenever the door is unlocked by using a keypad, Telegram, or face recognition. The Arduino drives the servo motor and moves it out of the locked position and to the unlocked position. After the door is unlocked, the hall effect sensor starts monitoring the magnetic state of the door. A magnet is put close to the opening of the door or lock. When the hall effect sensor spends the required time set in the code, the system is aware that the door has returned to the correct closed position.

-Once the magnet has been detected by the hall effect sensor over the preset time, the Arduino will restart rotation of the servo motor to the locked position. The LCD shows that the door is locked, and system goes back to its original locked position. The system will then delay until the next unlock request comes. In the entire process, the LCD screen shows some crucial messages, including; Doors Locked, Enter Password, Wrong password, and Doors Unlocked.

- **Challenges:**

  - ESP32 and Arduino: Proper connection.

  - Handling voltage differences

  - Face recognition accuracy

  - WiFi connection issues

  - Debugging communication signals

4. Sustainability & Impact

- **Sustainability:** Low power components are used in the system and the use of hard key is minimized. It may be applicable in smart homes to be efficient in using energy.
- **Impact:**
  - Improves security
  - Provides remote monitoring
  - Can be used in homes and offices.
- **Future Work:**
  - Add fingerprint sensor
  - Add mobile app
  - Increase the accuracy of face recognition.
- **Limitations:**
  - Depends on WiFi
  - Face recognition can also not work in low light.
  - Hardware complexity

5. Results & Discussion

This Smart Door Lock System based on Arduino and ESP32-CAM also was successfully tested and had all of the modules working correctly. The password system with a keypad unlocked the door appropriately and rejection to the incorrect inputs. The face recognition system was able to detect and recognize registered users and trigger the unlocking process. The control via Telegram also functioned correctly, supporting remote control and capturing of images, although this was only responsive on the internet connection. The servo motor was able to provide precision in lock-out and lock-in of the door and the hall effect sensor was able to detect reliably whether the door is open or closed. Generally, the system met its goals as it delivered a multi-layer secure access system.

Table: Experimental Results and Performance Analysis

Test Case	Input	Expected Output	Status
Keypad Access	proper passwords in the form of 4x4.	Pressing on the servo unlocks it, LCD displays: Door Unlocked.	**Pass**
Security Alert	Incorrect password entry	LCD displays Wrong Password, ESP32 takes photo.	**Pass**
Face Recognition	Sign in front of camera.	Send Signal to Arduino, Door opens.	**Pass**
Remote Control	/unlock command via Telegram	ESP32 is activated by Arduino Pin 13.	**Pass**
Low-Light Recognition	Face in dark environment, registered.	Secure unlock	**Pass**

Auto-Lock Logic	Door closed (Magnet near sensor)	Hall effect sensor picks up magnet	**Pass**
System Feedback	Status change / POW ON.	Real time LCD updates	**Pass**

Fig: 1.Password unlock 2. Face unlock 3. Telegram unlock

6. Conclusion

In this project, it was possible to create a smart door lock system with the Arduino and ESP32-CAM. It features a combination of password, face recognition and remote control.

The system enhances security and convenience over the ordinary locks. It demonstrates how hardware, and software can be combined to create smart applications.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Fariha Tanha	22101296	Display with arduino and keypad with arduino setup, co-integrator, wiring	● Developed the primary Circuit connection of Arduino UNO. ● Setting up The 4x4 keypad was connected with the Arduino digital pins, D2 to D9. ● Added LCD messages, including: Enter Password, Door Locked, Door Unlocked and Wrong Password. ● Wired the 20x4 I2C LCD to A4 and A5 of Arduino. Distribution of 5V and GND rail on the breadboard was managed. ● Verified wiring continuity that is checked and fixed jumper connections.
Shamaila Sadat Niha	22301728	Lock Mechanism and sensor setup	● Tested and attached the micro servo motor to Arduino D10. ● Adjust and set the program of the servo in lock and unlock movement. Attached the hall effect sensor to Arduino D11. ● Unblocked door opening/closing detection using hall sensor and magnet. ● Assisted with the physical lock set up to demonstrate.

Pranto Roy	22301261	ESP32-CAM hardware and transistor interface	● Wired both the ESP32-CAM module to the 5V and GND rails. ● Ready ESP32-CAM program environment. Linked ESP32-CAM GPIO output pins to the switching circuit of the transistor. ● Assembled The BC547 transistor interface with 10k Ohm resistors. Setting up of camera pins and ESP32-CAM board settings. ● Installed and tried the code of the ESP32-CAM camera web server. Changes/adds to the output signal of the GPS when there is a recognition/control signal. ● Tested the ESP32-CAM output signals in the case of the ESP32-CAM correctly connected to the Arduino input side.
Nishat Tasnim	22301643	Power management, final integration and testing	● Common Ground Connection between Arduino and ESP32-CAM modules, LCD, servo and sensors. ● Stability of managed power supply of Arduino, ESP32-CAM, and servo motor. ● Verified full circuit having put together all modules. ● Input authenticated by keypad, LCD display, servo movement, hall sensor activation and ESP32CAM signal activation.

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